

Diabetes Risk & Patient Health Report

1. Introduction

Diabetes mellitus is a long-term metabolic dysfunction where the body fails to control blood glucose levels. The International Diabetes Federation (2021) reports over 537 million adults worldwide are living with diabetes, projected to reach 783 million by 2045. In Sri Lanka, the prevalence rate among adults is 10.3%, making it a major national healthcare challenge (Katulanda et al., 2012).

Early detection is critical. Timely diagnosis allows for lifestyle changes and medication that can delay or prevent severe complications such as cardiovascular disease, chronic kidney disease, and diabetic neuropathy. Machine learning offers a low-cost, rapid alternative to conventional diagnostics by identifying at-risk individuals from routine clinical measurements.

This project implements a complete machine learning pipeline to predict diabetes development among female patients in the Pima Indian Heritage study, covering data preprocessing, feature engineering, training and tuning four classification models, evaluation, and deployment as a working prediction application.

2. Literature Review

The Pima Indians Diabetes Dataset has been extensively studied in ML research. Kavakiotis et al. (2017) found that Support Vector Machines and ensemble models outperformed conventional statistical methods, and that preprocessing — especially handling missing values — had a major influence on model performance.

Zou et al. (2018) achieved the highest accuracy of 77.7% using a Random Forest on the same dataset, attributed to the ensemble vote reducing individual tree variance. Sisodia and Sisodia (2018) found Naive Bayes achieved 76.3% accuracy but noted accuracy was insufficient due to class imbalance.

Tigga and Garg (2020) showed that feature selection before model training enhanced classification performance, with Glucose, BMI, and Age consistently emerging as the most predictive features — which directly informed the feature engineering strategy in this project.

3. Problem Definition

The project develops a binary classification model to predict the presence of diabetes based on 8 diagnostic health measures. Binary classification was chosen as the target variable (Outcome) takes one of two discrete values: 1 (diabetic) or 0 (non-diabetic). A full machine learning pipeline was developed using the Pima Indians Diabetes Dataset.

4. Data Collection

The Pima Indians Diabetes Database was sourced from Kaggle, originally developed by the National Institute of Diabetes and Digestive and Kidney Diseases (Smith et al., 1988). All subjects are female, aged 21 and above, of Pima Indian heritage. The dataset contains 768 patient records, 8 input features, and one binary target variable.

Class distribution: 500 non-diabetic (65.1%) and 268 diabetic (34.9%).

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration (mg/dL) — 2-hour oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-hour serum insulin (mu U/ml)
BMI	Body Mass Index (kg/m2)
DiabetesPedigreeFunction	Score reflecting genetic predisposition to diabetes based on family history
Age	Age in years
Outcome	Target: 1 = diabetic, 0 = non-diabetic

Table 1: Dataset Features and Descriptions

5. Data Exploration and Preprocessing

5.1 Exploratory Data Analysis

Key findings from EDA included: a moderate class imbalance (65:35) in favour of non-diabetic cases; a strong positive correlation between Glucose and Outcome confirmed by a heatmap; and histograms revealing that BMI, Age, and Pregnancies distributions are right-skewed.

5.2 Handling Zero Values

Zero values in five columns — Glucose, Blood Pressure, SkinThickness, Insulin, and BMI — are biologically impossible and represent missing data. These were replaced with column medians. The median was chosen over the mean due to its robustness against extreme outliers (e.g., Insulin values of 846 mu U/ml).

Column	Zero Count	Action Taken
Glucose	5	Replaced with median (117.0)
Blood Pressure	35	Replaced with median (72.0)
Skin Thickness	227	Replaced with median (23.0)
Insulin	374	Replaced with median (30.5)

BMI	11	Replaced with median (32.0)
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Table 2: Zero Value Handling in Preprocessing

5.3 Data Split and Feature Scaling

All eight features were standardized using StandardScaler (mean = 0, standard deviation = 1). This was necessary because distance-based algorithms (KNN, SVM) are sensitive to feature magnitude. The dataset was stratified-split into 80% training (614 samples) and 20% test (154 samples) to preserve class proportions.

6. Feature Engineering

Feature selection was performed using SelectKBest with the f_classif scoring (ANOVA F-test) from scikit-learn. This measures the statistical association between each feature and the target variable. Higher F-scores indicate more discriminative features.

Rank	Feature	F-Score	Clinical Significance
1	Glucose	212.4	Primary clinical diagnostic marker for diabetes
2	BMI	71.8	Central obesity is a major modifiable risk factor
3	Age	46.1	Insulin resistance increases significantly with age
4	DiabetesPedigreeFunction	40.7	Quantifies hereditary predisposition to diabetes
5	Pregnancies	17.6	Gestational diabetes raises future Type 2 risk
6	Insulin	12.5	Lower — high zero-count reduces informativeness
7	BloodPressure	7.2	Weak — less discriminative for this dataset
8	SkinThickness	6.1	Weakest — high missing rate reduces reliability

Table 3: Feature Importance Scores (SelectKBest)

SelectKBest was chosen over PCA to preserve original feature interpretability, which is essential in a medical setting. All eight features were retained for model training, with SelectKBest scores used to inform interpretation.

7. Model Training

Four supervised classification algorithms were trained on the processed training set:

Model	Category	Initial Parameters
Logistic Regression	Linear	max_iter=1000, random_state=42

K-Nearest Neighbours (KNN)	Distance-Based	n_neighbors=5
Support Vector Machine (SVM)	Distance-Based	kernel='rbf', probability=True
Random Forest	Ensemble	n_estimators=100, random_state=42

Table 4: Model Training Configuration

8. Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was used for hyperparameter tuning. Cross-validation provides a more reliable estimate of generalisation performance than a single train/test split (James et al., 2021). Random Forest and KNN were selected for tuning due to their sensitivity to key parameters.

Model	Parameter Tuned	Values Tested	Best Value
Random Forest	n_estimators	50, 100, 200	50
Random Forest	max_depth	None, 5, 10	5
Random Forest	min_samples_split	2, 5	2
KNN	n_neighbors	3, 5, 7, 9, 11	11
KNN	weights	uniform, distance	distance

Table 5: Hyperparameter Tuning Results

9. Model Evaluation

Each model was evaluated on the held-out test set using four metrics: Accuracy, Precision, Recall, and F1-Score. Recall was chosen as the primary metric because a False Negative (predicting non-diabetic for a diabetic patient) carries a much higher clinical risk than a False Positive.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.7013	0.5870	0.5000	0.5400
KNN (tuned)	0.7338	0.6327	0.5741	0.6019
SVM (RBF)	0.7338	0.6444	0.5370	0.5859
Random Forest (tuned)	0.7338	0.6585	0.5000	0.5694

Table 6: Model Evaluation Results on Test Set

10. Model Comparison & Selection

Random Forest (tuned) was selected as the best model. It achieved the highest Precision (0.6585) among all models and tied for the highest Accuracy (0.7338) alongside KNN and SVM. Key reasons for selection:

- Built-in feature importance scores allow identification of which diagnostic measurements most strongly predict diabetes — essential for clinical interpretability.
- Ensemble voting across 50 trees reduces variance and prevents overfitting that affects single-tree models.
- While KNN achieved marginally higher Recall (0.5741 vs 0.5000), Random Forest's clinical explainability was the deciding factor.
- Logistic Regression served as an effective baseline confirming linear structure in the data.
- SVM performed competitively but offered no predictability advantage over Random Forest.

11. Simple Deployment

The trained Random Forest model (`best_rf`) and the fitted `StandardScaler` were serialized to disk using Python's pickle library as `best_model.pkl` and `scaler.pkl`. A command-line prediction application was developed that:

- Loads both saved files at runtime.
- Accepts the eight diagnostic measurements as user input.
- Scales the input with the stored scaler to ensure consistency with training distributions.
- Outputs the predicted class (diabetic/non-diabetic) and probability of diabetes.

Saving the scaler alongside the model is critical — applying a different scaler would alter feature distributions and produce invalid predictions. This deployment follows the standard ML lifecycle: train once, deploy and predict many times without retraining.

12. Interpretation and Insights

Feature importance extracted from the tuned Random Forest ranked Glucose first, consistent with its central role in clinical diabetes diagnosis (American Diabetes Association, 2023). BMI and Age ranked second and third, confirming that lifestyle and demographic factors are powerful and accessible screening variables.

DiabetesPedigreeFunction ranked fourth, confirming the well-established hereditary component of Type 2 diabetes. Insulin and SkinThickness ranked lowest, partly due to high missing rates (48.7% and 29.6% respectively), which limited their informative content after imputation.

Two recommendations arise from these findings:

- Glucose measurement should be the initial screening test, given its strongly predictive nature.
- Data collection procedures should be improved to ensure Insulin and SkinThickness values are consistently recorded.

13. Reflection

13.1 What Worked Well

The pipeline was completed successfully. GridSearchCV effectively identified optimal hyperparameters without manual trial-and-error. Replacing impossible zero values with column medians had a significant positive effect on model performance. Stratified splitting preserved class proportions in both training and test sets, producing evaluation metrics representative of the actual class distribution.

13.2 Challenges Faced

The main challenge was recognizing that zero values represented missing data rather than real measurements — a domain knowledge issue not apparent from the data alone. Class imbalance (65:35) meant accuracy was a misleading metric; models maximizing accuracy might perform poorly on the minority (diabetic) class, requiring prioritization of Recall and F1-Score.

13.3 Possible Improvements

- Apply SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic diabetic records and improve Recall for the minority class (Chawla et al., 2002).
- Use a larger, more demographically diverse dataset to enhance model generalization.
- Compare gradient boosting algorithms (XGBoost, LightGBM) against Random Forest.
- Develop a web-based interface using Flask or Streamlit for accessibility by non-technical healthcare workers.

13.4 Key Learnings

This project validated that data quality and preprocessing have a greater impact on model performance than algorithm selection alone. The 'garbage in, garbage out' principle was demonstrated clearly: uncorrected zero values degraded model metrics significantly. It also reinforced that evaluation metric selection must be guided by the real-world consequences of each error type — in medical screening, reducing False Negatives takes clear precedence over maximizing raw accuracy.

14. Conclusion

This project successfully implemented a complete machine learning pipeline to predict diabetes using the Pima Indians Diabetes Dataset. Four classification models were trained, tuned, and evaluated — Logistic Regression, KNN, SVM, and Random Forest. After systematic comparison using clinically relevant metrics, Random Forest (tuned) was selected as the best model, achieving the highest Precision (0.6585) and providing built-in feature importance scores essential for clinical interpretability.

The project demonstrated that preprocessing decisions — particularly treating biologically impossible zero values and appropriate feature scaling — influenced model performance more than algorithm selection alone. Feature importance analysis confirmed that Glucose, BMI, and Age are the strongest predictors, consistent with established medical knowledge.

The deployed prediction application demonstrates that machine learning can serve as a cost-effective screening tool. Future work should focus on addressing class imbalance with SMOTE, exploring gradient boosting algorithms, and developing a web-based interface for clinical deployment.

15. References

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